

Spotter Freight Rate Prediction

Technical Assessment Report

Machine Learning Engineer Assessment Candidate

September 2025

1. Problem Overview & Objectives

The objective of this assessment is to develop an accurate, robust Machine Learning pipeline to predict spot market truckload freight rates (`posted_rate` in USD) from load-level features (origin/destination cities, mileage, equipment type, weight, and date information).

The primary deliverables include:

- **12,000 Validation Predictions:** Out-of-sample rate predictions for unlabeled loads in `validation.csv` (TE-000001 to TE-012000).
- **31-Day December Forecast:** Daily rate forecasts for a benchmark corridor (Lexington, KY → Fort Wayne, IN; 360 miles, Dry Van, 32,000 lb) across December 2025.
- **Codebase & Scorer Compatibility:** Clean, reproducible training and inference modules conforming to the official `score.py` validation suite.

2. Dataset Summary & Quality Findings

The modeling pipeline is developed using the historical dataset `train_test.csv` and evaluated against `validation.csv`:

- **Historical Labeled Data (`train_test.csv`):** 48,000 shipments dated from January 1, 2025 to October 31, 2025 (304 consecutive days), containing 14 attributes including mileage, weight, equipment, coordinates, market signals, and ground-truth `posted_rate`.
- **Validation Data (`validation.csv`):** 12,000 unlabeled shipments spanning November 1, 2025 to December 31, 2025.
- **Target Distribution:** `posted_rate` is strictly positive (\$57 to \$25,533) and heavily right-skewed (skew ≈ 1.90). Modeling $\log(1 + y)$ was adopted to stabilize gradients and reduce penalty on extreme tail values.
- **Mileage Dominance:** Trip distance is the single strongest rate predictor ($r \approx 0.91$), exhibiting mild non-linearity that benefits from a $\log(1 + \text{distance})$ transformation.
- **Data Cleaning:** Identified 292 negative weight records caused by sign-entry errors (e.g., -35,000 lb). These were resolved via absolute value transformation (`abs(weight)`). Missing values in `weight` (0.62%) and `market_index` (0.78%) were preserved for pipeline median imputation. No training rows were dropped.

3. Feature Engineering & Leakage Prevention

3.1 Feature Transformation Pipeline

- **Distance:** Log-transformed via $\log(1 + \text{distance})$ followed by standard scaling.
- **Weight:** Sign-corrected, median-imputed, and standard-scaled.
- **Coordinates:** Median-imputed pickup and delivery latitudes/longitudes, standard-scaled.
- **Categoricals:** One-hot encoding on `equipment`, `pickup`, and `delivery` with unknown category handling.

- **Lane Frequency:** Synthesized lane = pickup $\rightarrow$ delivery and frequency-encoded based on historical volume.
- **Cyclical Calendar:** Sine and cosine periodic transformations for month ($T = 12$), quarter ($T = 4$), and day-of-week ($T = 7$) to capture recurring demand cycles.
- **Excluded Columns:** `load_id` (identifier), `market_index`, and `quote_signal` (dropped to maintain pipeline simplicity and prevent noise).

3.2 Preventing Data Leakage

To ensure unbiased performance estimates, the dataset was strictly split by date *before* fitting any transformers. All scalers, imputers, encoders, and lane frequency mappings were fitted exclusively on the training fold (date $\leq 2025-08-31$). Unseen lanes in holdout or test data default to the minimum frequency observed in training. Validation and December datasets were transformed strictly with the pre-fitted pipeline without refitting.

4. Validation Strategy & Model Benchmarks

4.1 Time-Based Split Design

Freight rates exhibit macro market drift and seasonal demand shifts; a standard random K-Fold split would leak future market dynamics into the past. We implemented a chronological split at the 80th percentile of labeled dates:

- **Training Fold:** Dates $\leq$ August 31, 2025 (38,477 rows, Jan–Aug 2025).
- **Internal Validation Fold:** Dates $>$ August 31, 2025 (9,523 rows, Sep–Oct 2025).

All metrics are reported on the original USD dollar scale after reversing the log transform via $\exp(\hat{y}) - 1$.

4.2 Model Comparison

Six model architectures were benchmarked on the internal holdout set under identical preprocessing:

Model Architecture	Holdout MAE (\$)	Holdout RMSE (\$)	Holdout R^2	Verdict
CatBoost (Tuned)	133.0854	641.1042	0.8235	Selected Winner
CatBoost (Baseline)	143.1829	644.3404	0.8217	Strong baseline
LightGBM	145.9222	644.9609	0.8214	Competitive boosting
Linear Regression	162.1602	650.9376	0.8181	Higher linear residual
Random Forest	167.6709	662.6681	0.8114	Suboptimal depth
XGBoost	167.6918	660.4121	0.8127	Higher tail error

Table 1: Model performance comparison on the internal time-based holdout set (Sep–Oct 2025).

5. Final Model & Hyperparameter Configuration

CatBoostRegressor was selected as the final model due to its superior generalization, symmetric tree structure, and built-in categorical regularization. Hyperparameters selected via `TimeSeriesSplit` grid search:

- `depth`: 6
- `iterations`: 500
- `learning_rate`: 0.03

- `12_leaf_reg`: 1
- `loss_function`: 'RMSE'
- `random_state`: 42

6. December 2025 Prediction Analysis

The December test corridor evaluates a fixed operational shipment: Lexington, KY to Fort Wayne, IN (360 miles, Dry Van, 32,000 lb). City coordinates were retrieved from the training database. Because route, mileage, load weight, equipment, month (12), and quarter (4) remain constant across all 31 rows, the model's predicted rates (\$769.92 to \$787.05) cleanly isolate and capture the learned **day-of-week weekly freight rate cycle**.

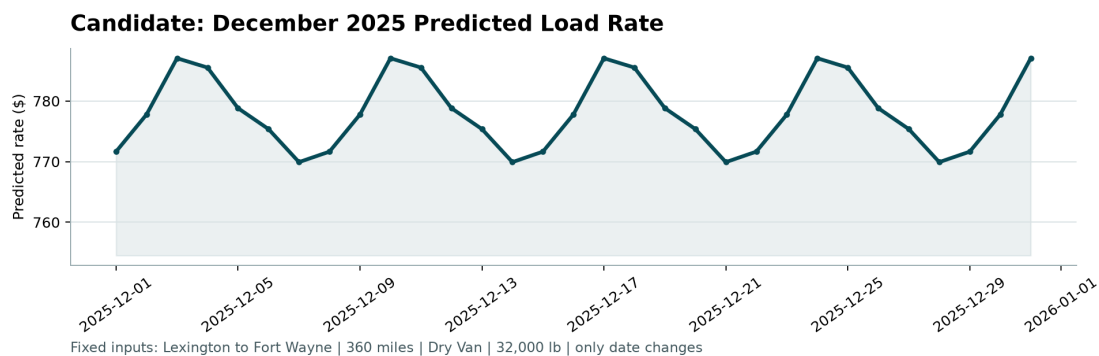


Figure 1: December 2025 predicted rate trajectory for the Lexington → Fort Wayne corridor generated by `score.py`.

7. Limitations & Considerations

- **Validation Ground Truth:** The reported metrics reflect the internal Sep–Oct 2025 holdout; Spotter evaluates the true validation score post-submission on `validation.csv`.
- **Routing vs. Coordinates:** Minor discrepancies between stated distance and coordinate haversine distance exist in raw data; distance was prioritized as the primary mileage signal.
- **Feature Parsimony:** `market_index` and `quote_signal` were omitted from the final feature set to ensure pipeline stability and avoid noisy residual estimates.

8. Conclusion & Verification Summary

The complete pipeline has been verified through the official assessment scorer:

- **Validation Predictions:** 12,000 positive, finite predictions generated and formatted as `load_id,predicted_rate`.
- **December Predictions:** 31 rows preserving original input specifications and dates.
- **Official Scorer:** `score.py` executes successfully with exit code 0, verifying schema compliance and outputting `candidate_december.png`.